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Formalizing Ecosafety: A Data-Driven Framework for Predicting Substrate Degradation, Battery Risks, and System-Level Control in Cyboquatic Networks

Quantitative and Qualitative Analysis of Flow Regime Impacts on Substrate Degradation

The investigation into the impact of varying flow regimes on channel substrates requires a dual-mode analysis that integrates quantitative modeling of microplastic shedding rates with a qualitative understanding of the underlying degradation mechanisms [13] [75]. This approach ensures that the resulting risk models are not only predictive but also interpretable and tunable, adhering to the core principles of the Cyboquatic ecosafety grammar [5] [9]. The primary distinction lies between laminar, transitional, and turbulent flow regimes, each imposing unique physical stresses that drive distinct degradation pathways. Laminar flow, characterized by smooth, parallel streamlines and low shear stress, primarily facilitates chemical and enzymatic surface hydrolysis [75]. In this regime, the degradation process is slower, dominated by intrinsic material properties rather than external forces. Consequently, microplastic shedding is minimal and consists mainly of slow surface pitting and erosion [13]. This condition serves as a proxy for measuring the material's inherent chemical stability, which can be quantified using a chemical rate constant (`k_chem`). Transitional and turbulent flows, common in natural and engineered canals, introduce significant mechanical forces. These regimes are defined by higher Reynolds numbers (Re), which characterize the transition from orderly laminar flow to chaotic, eddying turbulent flow [26]. Turbulent flow introduces substantial wall shear stress (`τ_wall`), which adds abrasion, impact, and erosion to the degradation process [25] [54]. This combination of chemical and mechanical stress accelerates bulk mass loss and gives rise to a "fragmentation-first" mode, where the substrate's stiffness and modulus decay before a significant amount of total mass is lost [23] [64]. This phase can generate an early pulse of particulate residue, creating a separate and critical risk coordinate (`rmicro`) that must be monitored independently of bulk degradation metrics like `t90` [109]. To model this, a fragmentation kernel (`J_micro(t, Re)`) can be developed to

predict microfragment flux as a function of hydraulic parameters, directly feeding the `rmicro` coordinate [21].

The qualitative understanding of degradation mechanisms—such as crack growth, fibrillation, and hydrolysis—is crucial for interpreting the outputs of these quantitative models [20] [86]. For instance, hydrolytic crack growth occurs when polymer bonds react with water molecules under tensile stress, leading to brittle fracture [20]. This mechanism is particularly relevant in moist environments and can be exacerbated by cyclic loading and fatigue [63] [83]. Fibrillation, the formation of fine filaments from the polymer matrix, contributes to microplastic shedding and can be influenced by both mechanical stress and chemical exposure [74]. By mapping these mechanisms to the observed behavior under different flow regimes, the risk kernels become more transparent. For example, a high `rmicro` value in a laminar flow test would indicate an intrinsically brittle or fibrillating material, while a similar value in a turbulent flow test might point to mechanical fragility under shear stress. This interpretability allows engineers to tune material formulations or deployment strategies—for instance, avoiding mechanically fragile substrates in high-shear reaches of a canal network even if their bulk degradation time (`t90`) appears acceptable [23] [64]. The integration of these concepts is facilitated by the structured data schema, where each experimental run explicitly records the flow regime (`laminar`, `transitional`, `turbulent`), representative Reynolds number (`Re_min`, `Re_max`), and wall shear stress (`tau_wall_pa`) alongside degradation metrics like `t90_days_lab` and `j_micro_max_mg_m2_d` [25] [26]. This explicit linkage transforms flow regime from a mere variable into a structured driver of the existing risk coordinates (`rt90`, `rmicro`, `rtox`), which collectively feed into the materials slice of the Lyapunov residual `Vt`. The system is designed to gate any substrate-flow regime combination that produces either a long `t90` or a high `microresidue` count, thereby enforcing the principle that fast-degrading materials must also be low-shedding to be deployable [96].

Risk Coordinate	Description	Primary Driver in Laminar Flow	Primary Driver in Turbulent Flow
`rt90`	Normalized time to 90% mass loss.	Chemical/Enzymatic Hydrolysis [75].	Mechanical Abrasion and Erosion [23].
`rmicro`	Normalized microplastic/ microfragment risk.	Slow Surface Pitting [13].	Fragmentation, Shear-Induced Breakage [109].
`rPFASresid`	Normalized risk from PFAS residues.	Leaching from bulk matrix.	Enhanced leaching from exposed internal structure.
`rleachCEC`	Normalized risk from other CECs.	Diffusive leaching.	Convective leaching enhanced by surface damage.

This dual-mode approach, combining rigorous quantitative measurement with clear qualitative mechanistic interpretation, provides a robust foundation for assessing substrate performance. It enables the development of targeted testing protocols and informed decision-making, ensuring that the selection of materials is optimized not just for longevity but also for minimizing the release of harmful particulates into aquatic ecosystems. The ongoing refinement of these models through systematic experimentation across the full range of flow regimes will further tighten the associated corridor bands, reducing uncertainty and enhancing the overall safety and efficacy of Cyboquatic systems [54] [80].

Standardized Accelerated Aging Protocols for Predictive Performance Forecasting

The development of an accelerated aging protocol to reliably predict the multi-year `t90_days` for novel substrate formulations represents a critical step in de-risking field deployments. The strategic directive prioritizes the establishment of standardized, Phoenix-anchored test conditions that are reproducible across laboratories over the immediate creation of complex predictive algorithms [5] [9]. This phased approach mitigates the significant danger of "blind extrapolation," where accelerated tests are misinterpreted, leading to unreliable lifetime predictions [14]. The foundation of this protocol is the creation of a canonical set of accelerated conditions, defined by specific, repeatable parameters such as elevated temperature (e.g., 30–40 °C), controlled dissolved oxygen (DO), standardized biofilm inoculum, and defined shear bands [38] [52]. By anchoring these protocols to the specific environmental conditions of Phoenix, Arizona—including its characteristic diurnal and seasonal temperature profiles—the resulting datasets will be directly relevant to the target deployment environment [6] [7]. This standardization aligns with established international practices like ISO and OECD guidelines, which emphasize the importance of reproducibility in testing polymers and chemicals [48] [50]. For instance, the ISO 4892 series specifies methods for exposing plastics to laboratory light sources to simulate weathering effects [38] [120], providing a precedent for defining controlled, albeit intensified, environmental stressors.

Once a robust dataset is generated from these standardized, reproducible tests, predictive translation algorithms can be developed as a secondary layer [117]. These algorithms would leverage established physical and chemical principles, such as the Arrhenius equation or Q10-based relationships, to extrapolate the measured lab kinetics (`k_lab`) to real-world

field conditions (`k_field`) [34] [56]. The Arrhenius law describes the temperature dependence of reaction rates, allowing the acceleration factor at a given lab temperature to be calculated relative to an average field temperature [35] [58]. However, this prediction is not treated as a standalone result. Instead, it is always anchored by the baseline data derived from the Phoenix-anchored protocol. The final, field-relevant metric, `t90_field_days`, is computed by integrating the actual, time-varying temperature profile of the target canal reach against the fitted kinetic surface (`k(T, shear)`) [56] [58]. This correction for real-world temperature variability is essential, as a single constant-temperature lab `t90` systematically misrepresents field degradation; warmer periods accelerate decay while cooler periods slow it down, a nuance captured by integrating the temperature history [5] [9]. The entire process is encapsulated within the `BiodegradableSubstrateChannelKinetics2026v1.aln` shard, which stores both the raw lab-derived metrics (`t90_days_lab`, `k_chem_ref_day`) and the climate-corrected, field-predicted values (`t90_days_canal_est`) [25] [26]. This ensures that the most ecologically relevant metric is used for deployment decisions. To enforce quality and reliability, a "no corridor, no build" policy is implemented at the protocol level: a substrate formulation only earns a production-grade `t90_field_days` value if its accelerated results and at least one confirmatory, long-term, lower-stress test fit the same kinetic model within a predefined tolerance [51] [119]. If not, the risk coordinates remain high, and the material stays in a research-only lane. This methodology builds confidence in the data, moving risk discovery forward in time and suppressing the deployment of materials prone to long-lived residues or excessive microplastic shedding before they ever enter a field pilot [12] [36].

Balanced Risk Assessment Framework for Battery Chemistries Across Their Lifecycle

A balanced risk weighting criteria for battery chemistries necessitates a sophisticated, multi-layered framework that integrates environmental impacts, supply chain considerations, and end-of-life (EoL) pathways into a cohesive decision-making structure [19] [42]. The user's directive mandates a hierarchical approach where environmental and EoL factors serve as hard-gated coordinates, meaning they cannot be overridden by logistical or economic considerations. Supply chain risk, conversely, primarily functions as a knowledge factor (K) and influences deployment lanes (`lane=PROD`), shaping the practicality and scalability of a technology without compromising fundamental safety and sustainability principles [40] [41]. This creates a robust system that rewards technologies

with benign footprints while acknowledging real-world implementation challenges. Environmental and EoL impacts are formalized as normalized risk coordinates (rx) that are aggregated into the Lyapunov residual Vt. Key coordinates include r_tox_leachate (derived from OECD-style aquatic toxicity tests on shredded cells), r_PFAS_like_additives (if applicable), and rcarbon_e_eol (the end-of-life carbon impact) [49] [50]. Similarly, the energy plane risk renergy incorporates degradation by weighting carbon per MWh by a time-varying round-trip efficiency η(t); a slow-degrading cell will carry a higher effective renergy trajectory than a well-managed one [19] [44]. Any chemistry whose rmaterials_battery or rcarbon_energy coordinates violate the established safe/gold/hard corridor bands fails the corridorpresent check and is barred from production deployment, regardless of its supply chain attributes [69] [107].

Supply chain considerations, such as resource criticality, geopolitical concentration, and recycling infrastructure maturity, do not operate at this hard-gated level. Instead, they inform the knowledge factor (K) and influence the choice of deployment lane [81] [92]. For example, a sodium-ion (Na-ion) battery may have a lower materials risk (rmaterials_battery) due to its use of abundant elements, making it a strong candidate for production [45]. An NMC (Nickel Manganese Cobalt) battery with a lower carbon footprint but higher critical metal content might occupy a middle ground, while a new, unproven chemistry with a problematic supply chain might be relegated to the RESEARCH lane until market conditions improve [46]. This tiered approach allows for a pragmatic yet principled technology selection process. The refinement of rcarbon_energy is central to this framework, as it moves beyond simple manufacturing emissions to encompass the full lifecycle. The net carbon intensity is calculated as a composite score:

$$r_{carbon_energy} = K_{carbon}(M_{manufact} + M_{use} + M_{EoL})$$

where M_use is adjusted by the round-trip efficiency η(t), and M_EoL is a scenario-dependent term reflecting recycling, repurposing, or disposal [44] [46]. Recycling offers a credit by displacing primary mining, while landfilling incurs a penalty [42]. This comprehensive rcarbon_energy score, along with rmaterials_battery, feeds directly into the KER triad, enabling a direct, quantitative comparison between Na-ion, LFP (Lithium Iron Phosphate), and NMC chemistries. LFP, for instance, often has low cobalt/nickel content, giving it a favorable rmaterials_battery, while Na-ion benefits from low criticality but may require careful evaluation of novel components for new leachate risks [45] [70]. The entire framework is encoded in the EnergyBankModuleEcosafety2026v1.aln shard, which defines sub-coordinates,

corridor bands, and weights for each chemistry, ensuring consistent application of the risk assessment logic [19].

Battery Chemistry	Typical `rmaterials_battery` Drivers	Typical `rcarbon_energy` Drivers	Typical Supply Chain Considerations
Na-ion	Low criticality of Na, Fe, P. Potential novelty of some precursors/binders [45].	Generally lower GWP due to less energy-intensive processing and no Co/Ni. High recycling potential [70].	Abundant raw materials, but mature recycling infrastructure is less developed than for Li-ion.
LFP	Low Co/Ni content, but sometimes higher mass/kWh. Lower toxicity than NMC [19].	Moderate GWP. Well-established manufacturing. High recycling rates for Fe/P [42].	Mature supply chain with established recycling. Less exposure to volatile Co/Ni markets.
NMC	Higher content of critical/coarse metals (Ni, Co). More complex recycling process [19].	Higher GWP due to energy-intensive Ni processing and solvent use.	Concentrated supply chains for Co and Ni. Geopolitical risks and ethical sourcing concerns [81].

This balanced framework provides a defensible, data-driven method for selecting battery technologies that are not only efficient and cost-effective but also environmentally responsible and resilient. By hard-gating environmental and EoL factors, it ensures that short-term convenience does not lead to long-term ecological harm, while using supply chain data to guide realistic and sustainable deployment strategies.

Expanding the Kinetic Shard for Additives, Biofilms, and Sorption Dynamics

To create a truly comprehensive and ecologically aware framework, the kinetic shard must be expanded to explicitly account for the fate of additives, the dynamic role of biofilms, and the substrate's capacity to act as a passive sampler for environmental contaminants [112][115]. These additions move beyond simple bulk degradation metrics to capture a richer, more nuanced picture of a substrate's environmental interaction, thereby tightening the associated risk corridors and improving the fidelity of the KER triad. First, additives such as plasticizers, stabilizers, and oligomers are often not part of the primary polymer backbone and can leach out over time, posing their own ecotoxicological risks [115]. The current `rtox` coordinate captures this broadly, but treating them as first-class analyte families provides greater precision. This involves extending the `BiodegradableSubstrateChannelKinetics2026v1.aln` shard with columns for additive-specific analytes (e.g., ATBC, glycerol, BD) and their measured concentrations over time in standardized tests [109]. For each additive family, a dedicated normalized risk

coordinate (`r_additive_i`) is defined based on its concentration relative to ecotoxicological thresholds (e.g., NOEC/EC50) [73]. These additive-specific risks are then explicitly aggregated into the parent `rtox` coordinate via a quadratic form:

$$r_{tox}^2 = \sum_i w_{add,i} r_{add,i}^2 + \cdots$$

This makes additives visible in both the lab and field shards, ensuring that recipes which pass bulk `t90` tests but exhibit persistent or spiking ecotoxicity from additives are correctly flagged and rejected for deployment [72]. This closes a major gap between "bulk biodegradation looks fine" and "additive leachate is ecotoxic."

Second, biofilms are recognized as the "engine" of both degradation and contaminant mobilization, yet their state is currently an implicit variable [54] [55]. To make them explicit, the kinetic shard should include optional fields for biofilm characterization, such as biomass (g/m²), dominant microbial guilds (from 16S rRNA sequencing), and extracellular polymeric substance (EPS) characteristics [55]. This leads to the definition of a normalized `rbiofilm_state` coordinate with safe/gold/hard bands tied to desired functionality (e.g., stable, diverse communities) and ecotoxic risk (e.g., avoiding blooms of problematic consortia) [27]. Biofilm dynamics directly influence leaching rates and PFAS mobility, so tracking `rbiofilm_pfas` becomes critical [10] [11]. A biofilm sorption capacity (`K_d`) scales the potential for both capturing and releasing contaminants, making `rbiofilm` an essential explanatory variable in the kinetic residual `V_materials` [66] [76]. Finally, substrates can accumulate contaminants from the water column, effectively acting as passive samplers [106][108]. This sorption capacity must be explicitly modeled. The kinetic shard should include sorption isotherm parameters (e.g., `sorption_kd_L_kg`) and an "optional passive sampler" flag [106]. When enabled, the system tracks the mass of contaminants accumulated in the substrate (`Msorb`) and computes a corresponding risk coordinate (`r_sorbed_mass`). This coordinate serves two purposes: it acts as evidence of basin-scale contamination (a positive eco-impact) and as a stored inventory that must be managed before structural disintegration remobilizes it [108]. Governance rules can be implemented to treat `rsorb_*` as a deployment gate, requiring mandatory removal or regeneration once the sorbed mass crosses gold bands to prevent later uncontrolled release [69]. These expansions transform the shard from a simple record of degradation into a powerful diagnostic tool that illuminates multiple pathways of environmental interaction, all of which are made visible, normalized, and enforceable through the ecosafety grammar.

Modeling Structural Integrity and Contaminant Fate in Channel Substrates

Beyond bulk mass loss, a critical aspect of substrate degradation is the "structural disintegration" phase, which can precede significant mass loss but trigger a pulse of particulate residue [23] [64]. This phenomenon, involving processes like cracking, embrittlement, and delamination, must be modeled separately from traditional kinetics to accurately predict the generation of microfragments. The proposed framework addresses this by introducing a second state variable or coordinate to represent substrate integrity, distinct from the mass fraction remaining [23] [64]. This could be a normalized stiffness, a damage variable (`D(t)`), or a fracture index, governed by its own kinetic law. This integrity variable would be driven by factors like hydrolysis, fatigue from cyclic loading, and penetration by biofilms or roots [63] [83]. Disintegration is modeled as an event that occurs when this integrity variable crosses a predefined threshold, even if the gravimetric mass loss is still modest (e.g., <20%) [64]. This event would then seed a microfragment flux (`J_micro`), which decays as those fragments themselves biodegrade or settle [109]. To implement this, a two-channel kinetic model is required for each substrate, with lab tests designed to monitor both bulk mass loss and the structural integrity variable simultaneously [118]. The resulting `rstruct_int` coordinate, representing the deviation from a safe structural state, would be fed into the materials residual `V_materials` [93]. This allows for the gating of materials that become structurally unsound prematurely. The fragmentation rate and resulting `rmicro` are coupled to hydrodynamic and shear kernels, making the risk of particle release an explicit function of both material state and canal hydraulics, not just time [23] [64]. This enables the creation of a "disintegration event" diagnostic frame that predicts and logs expected pulses of particulates under specific loading scenarios, allowing for operational derating or blocking in sensitive reaches [118].

The fate of contaminants, both those leached from the substrate itself and those sorbed onto it, is another crucial component of the framework [112][115]. The CEIM mass-balance machinery already handles contaminant loads, but it must be adapted to account for the substrate as a dynamic sink. The substrate's sorption capacity is governed by a partition coefficient (`K_d`), and the accumulation of contaminants like PFAS and other CECs follows a simple ODE: $\mathrm{d}M_{\mathrm{sorb}}/\mathrm{d}t = k_{\mathrm{ads}} C_{\mathrm{water}} - k_{\mathrm{des}} M_{\mathrm{sorb}}$ [106]. The kinetic shard includes `sorption_kd_L_kg` as a key parameter [106]. As the substrate degrades, this stored inventory of contaminants becomes a source term that must be accounted for in the basin-level mass balance, preventing an underestimation of the total pollutant load [66]. The framework uses normalized risk coordinates (`rPFASresid`, `rleachCEC`) to

quantify the risk from both leached and sorbed fractions, ensuring that materials which concentrate toxins are not deployed in sensitive areas [69]. Furthermore, chronic, low-dose exposure to the complex mixture of leachates from multiple Cyboquatic nodes within a single basin is addressed by shifting the controlled system from individual nodes to the basin as a whole [6] [7]. Each node's contribution is a time-series of risk vectors (`rPFAS`, `rCEC`, `rmicro`, etc.), which are summed across the network using CEIM mass-balance kernels [66]. This generates basin-scale residuals (`Vaquatic`, `Vworm`, etc.) that are subject to hard caps (e.g., `Vspecies ≤ 0.10`) and monotonic constraints (`V_species,t+1 ≤ V_species,t` under allowed changes) [96]. This ensures that cumulative, low-dose mixture effects are managed at the ecologically relevant scale, bounding subtle impacts like shifts in microbiome composition or chronic microfragment ingestion [111][113]. The system thus evolves from a collection of individually-safe nodes to a network of nodes operating within a collectively-safe basin-wide constraint.

Integrated Framework for Energy Banks and Canal-Tied PV Systems

The integration of Cyboquatic energy banks with "canal-top solar" (PV) systems presents a synergistic opportunity to enhance both the technical performance and ecosafety of the overall system [53] [78]. The proposed framework reflects these co-benefits by expanding the energy bank's risk coordinates and modifying the surrounding planes accordingly, all within the existing ecosafety grammar. The primary co-benefit stems from the thermal regulation provided by the canal environment. Water has a high heat capacity, which moderates temperature fluctuations for both the canal lining and the PV panels mounted above it [67]. Cooler operating temperatures for PV modules significantly increase their electrical conversion efficiency, while cooler temperatures for battery energy storage systems (BESS) reduce thermal stress, lower self-discharge rates, and extend cycle life [30]. In the energy plane (`renergy`), this translates to lower thermal stress coordinates (`rthermal`) and potentially lower round-trip losses, leading to a higher effective efficiency (`η_eff`). This improvement in operational efficiency directly lowers the `renergy` risk coordinate, as less energy is wasted during charge/discharge cycles [56]. The energy bank's shard schema can be extended to include fields for `placement_lane` (e.g., `CANAL_TOP_PV`) and correlated temperature corridors, which are used to adjust the `renergy` and `rcarbon_energy` coordinates downwards compared to land-based installations [28] [31].

From a hydraulic perspective, the presence of PV support structures introduces new design considerations. While shading from the panels reduces evaporation and thermal stratification, which are generally beneficial, the structures themselves can alter wind shear and cause local head losses or surcharge if poorly designed [28] [31]. The `ArtemisCyboquaticNodePlacement2026v1` shard is extended to include host infrastructure IDs for PV assets, and hydraulic kernels are used to compute a revised `rhydraulic_design` that accounts for these new obstructions [28]. Non-actuating frames like `HydraulicDecayFrame` can monitor how PV shading affects water velocity and sedimentation patterns, updating the `rhydraulic` coordinate accordingly [30]. The most significant benefit, however, is realized in the carbon plane. Local PV generation provides zero- or low-intensity electricity for charging the BESS, drastically reducing reliance on the fossil-fuel-dominated grid [39]. This is reflected in the `CarbonCorridor` kernel, which routes the PV output as having a near-zero carbon intensity (`grid_gco2_kwh_node ≈ 0`) [39]. The result is a dramatic reduction in the `rcarbon_energy` coordinate for the energy bank, pushing it towards or below zero, especially when combined with high-efficiency operation [44] [46]. This turns the site into a net carbon-negative buffer, a powerful eco-benefit that is explicitly captured in the risk assessment.

Finally, the Lyapunov residual `vt` can be leveraged as a real-time control signal to optimize the energy bank's operations for both grid services and ecosafety [96]. Since `Vt` is a scalar measure of aggregate ecosystem risk, it can serve as a veto or steering mechanism for control actions. A SafeController proposing a charge/discharge sequence would have its actions evaluated by the non-actuating `ecosafety-core` crate, which calculates the change in `Vt` (`ΔVt`) [96]. Steps that would increase `Vt`—such as a fast-charge that pushes the battery into a high-thermal-stress regime (raising `renergy`) or a discharge timed to coincide with a period of high background toxicity (`rtox`)—would be rejected or derated by the `safestep` gate [81] [96]. Conversely, actions that provide grid services (e.g., frequency regulation) while maintaining or lowering `Vt` (e.g., using surplus PV power) are accepted [30]. This transforms `Vt` from a static diagnostic into a dynamic "eco-Hamiltonian" that guides scheduling policies in real time, favoring low-carbon, low-stress operating windows and discouraging ecologically sensitive actions [81] [96]. The entire framework, from the PV-cooled battery to the `Vt`-guided controller, remains strictly within the existing ecosafety grammar, demonstrating its adaptability to complex, interconnected systems while rigorously enforcing environmental safeguards.

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